

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			award (1) for either of following energy of the <i>d</i> -electrons are similar successive ionisation energies are similar	1			1		
	(b)	(i)		blue (1) the frequency of light that is <u>not absorbed</u> is in this region (1)		2		2		
		(ii)	I	a (small) molecule/ion with a lone pair (that can bond to a transition metal ion)	1			1		
			II	award (1) for any of following lone pair from ligand combines with empty orbital lone pair from ligand can combine with electron deficient transition metal atom sharing of a pair of electrons which are both from ligand forms a coordinate bond (1)	2			2		
	(c)	(i)		$2\text{I}^- \rightleftharpoons \text{I}_2 + 2\text{e}^-$		1		1		
		(ii)		$\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightleftharpoons \text{VO}^{2+} + \text{H}_2\text{O}$		1		1		